

Metempsychosis in Media, Performance, and Sleep Choreography

Yiou Wang

Harvard University Graduate School of Design | Mixanthropy Studio, USA
yiou@mixanthropy.com

ABSTRACT

Does oneiric states of consciousness drive digital and analog characters? Re-inventing mythological archetype of one's dreamed self with augmented cognition and myth-informed mind-body relations, this talk explores metempsychosis in the dialogue between mythological narrative and technocultural imagination, describes the development process of the sleeper's characterization and sleep choreography through the media x performance installation "Morphai," as part of MITACS Fellowship and Sociability of Sleep, and discusses several novel approaches to sleep choreography fusing technology and body art. The notion "morphai" comes from the human shapes that the Greek god of dreams, Morpheus, sends to people when they fall asleep. Metempsychosis refers to the transmigration of a soul from one body to another which, in terms of body art and performance art, is externalized in movement. "Morphai" syncs the sleep performer's dreamed self to two characters – the augmented avatar and the linework figure from moving inside a phosphorescent costume. Featuring the performance by artists Dayna McLeod and Laurie-Anne Gosselin, two Morphai dance to the score of the sleep performer's own motion, neurological and physical, as an extension to the self in a continuous landscape of consciousness. As the digital Morphai is animated through biosensor data in sleep, the performer is dressed in a phosphorescent costume that blurs out their day identity but accentuates their night form, as a ritualized investigation of if and how sleep can be performed, fusing digital media and performance. The work "Morphai" is exhibited at Sociability of Sleep's culmination exhibition "InSomnolence" curated by Marianne Cloutier, Alanna Thain and Aleksandra Kaminska, from June to July, 2023, in Tio'tia ke/Montréal.

ACM Reference Format:

Yiou Wang. 2023. Metempsychosis in Media, Performance, and Sleep Choreography. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595438>

1 INTRODUCTION

Media art historian Lev Manovich posits that today's best art is produced in the fields called "body arts" [Manovich, 2023], as such fields both use technology in interesting ways and employ the human body as a portal and an opportunity. Taking it one step further, the most promising fields of innovation in arts today lie in the fusion between media and performance.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595438>

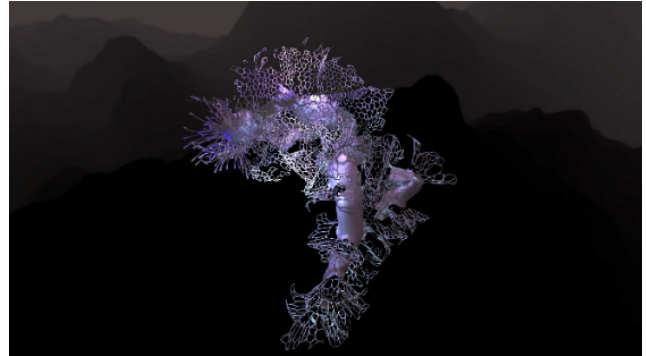


Figure 1: Dream avatar of a performer, with morphological transformations driven in the real time by EMG sensors placed on the sleep performer's body

This talk starts from the definition of metempsychosis – a philosophical term describing a soul's migration between bodies, a modern mirror reflection of which is motion capture retargeted animation. In the context of cyborg posthumanism and technology-enabled possibilities, the concept of metempsychosis evolves in an ongoing exchange between mythology and technology. It is not only how mind-body relationships in flux are narrated, but the techno-artistic research creation methods contribute to the development of the notion of metempsychosis. Sleep and dream are a largely underexplored domain of media arts, presumably because they are highly private in the modern time where individuals draw clear-cut lines between public and private. But sleep is interesting in that it encompasses a wider range of consciousness fluctuation and modes of fictive movements. Neuroscience findings indicate that the acting out of the fictive movements in dream has a noteworthy connection to the contents of dreams [Nielsen and Kuiken, 2013; Dresler et al., 2011], and flux consciousness in the sleep state progression [Voss et al., 2013]. Dream and wakefulness can engage in a two-way interaction. Dreams have the potential to be engineered, modified, and influenced by wakeful intentions via wearable devices such as Dormio [Haar Horowitz et al., 2020] just as an individual's wakeful emotion can be expressed and communicated socially such as in "Emotional Beasts" [Bernal and Maes, 2017]. In such context, sleep provides rich soil for emergent artistic methods of metempsychosis, not only because sleep is a body art, but also because it opens up aspects of fictive movements (dream movements that correspond to subjective dream experience) as well as movements across a wide landscape of consciousness not present in wakefulness.

2 SLEEP CHOREOGRAPHIES

“Morphai” syncs the sleep performer’s dreamed self into two characters – the first morphai is a digital avatar with procedural morphological transformations driven by EMG sensors placed on the live performer, as their externalized dreamed self; the second character is the phosphorescent glowing lines on the costume worn by the sleep performer while sleeping, making strange patterns with their dream movements.

Oneiric mind-body relationships informed by Greek myth of morphai is essentially a characterization of sleep, where the morphai is the character. It is more than representation since the character responds in the real time and hence gains agency. It interrogates what consciousness is and whether human consciousness can be combined with technical apparatus. The digital avatar’s movement is a dance of bio-signals designed to be highly sensitive to sub-threshold EMG activation. This pathway of sleep characterization represents a cyborg identity expression where the body has an apparatus which sends and receives signals and becomes aestheticized as an expressive mind-body resonance. The design of the avatar evokes vegetative and fungal growth to further characterize sleep as a series of states of alternative consciousness.

Although both deal with puppetry and metempsychosis in their unique ways, the costume characterization via costume is more visibly reminiscent of theatrical puppetry, where the performer erases their own identity but uses movement to create another identity. This way is what I call Analog Mocap, where the linework on the costume are analogous to the skeleton to animate. The performance video anonymizes the performer’s wakeful identity, leaving only the abstract linework/skeleton. One’s wakeful visual appearance gives away much information that subconsciously alter the way one is treated socially, but when I want to amplify the choreography of a combination of movements on a spectrum of fiction and consciousness, I do not want to distract the audience by those daytime identifiers and markers.

Metempsychosis is not a new practice; many Asian, African, and indigenous American cultures incorporate metempsychosis in their belief systems, especially when it involves the transition between wakeful life and dreamed life. Media research and arts investigating metempsychosis embrace a nonbinary thought where expression and identity are directed not by kind but by movement. To transmigrate a soul from a sleeper’s body to another involves animating the body to migrate into, a procedure that can be alienating the performer’s motion from their appearance. This alienation facilitates us to better focus on the movement expression.

3 FUTURE WORK

“Morphai” syncs the Sleep is a two-way portal. The installation exhibited at “InSomnolence” is only the start of a Morphai Universe. My future development of Morphai includes two major directions: the first one is scheduled to be delivered in the last quarter of 2023 so no details could be released yet, but it involves an in-depth use of sleep observation and documentation. The second future direction involves a two-way feedback loop. The Morphai Universe is a long-term project that will involve several branches and collaborations with multiple performers and actors.



Figure 2: Phosphorescent lines on the costume creates a reductive metempsychosis or Analog Mocap

ACKNOWLEDGMENTS

This research creation is funded by MITACS Fellowship 2022-2023. It was made possible by Sociability of Sleep project leads Alanna Thain and Aleksandra Kaminska, curator Marianne Cloutier, my wonderful colleagues and interesting interlocutors. Artists Dayna McLeod and Laurie-Anne Gosselin’s participation completed this project with tangible synergy. My friends including but not limited to Kévin Desfossés, Clement Curay offered their passion and time to assist and participate, in this stage of “Morphai” and the next.

REFERENCES

- G. Bernal and P. Maes. (2017). Emotional beasts. Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems. <https://doi.org/10.1145/3027063.3053207>
- M. Dresler, S. P. Koch, R. Wehrle, V. I. Spoormaker, F. Holsboer, A. Steiger, P. G. Sämann, H. Obrig, and M. Czisch. (2011). Dreamed movement elicits activation in the sensorimotor cortex. *Current Biology*, 21(21), 1833–1837. <https://doi.org/10.1016/j.cub.2011.09.029>
- A. Haar Horowitz, T. J. Cunningham, P. Maes, and R. Stickgold. (2020). Dormio: A targeted dream incubation device. *Consciousness and Cognition*, 83, 102938. <https://doi.org/10.1016/j.concog.2020.102938>
- R. Mallett. (2019). A Pilot Investigation into Brain-Computer Interface Use during A Lucid Dream. <https://doi.org/10.31219/osf.io/my3tq>
- L. Manovich. (2023, June 10). LinkedIn. <https://www.linkedin.com/feed/update/urn:li:activity:7073390064231206912/>
- T. Nielsen, and D. Kuiken. (2013). Relationships between non-pathological dream-enactment and Mirror Behaviors. *Consciousness and Cognition*, 22(3), 975–986. <https://doi.org/10.1016/j.concog.2013.06.005>
- U. Voss, K. Schermelleh-Engel, J. Windt, C. Frenzel, and A. Hobson. (2013). Measuring consciousness in dreams: The lucidity and consciousness in dreams scale. *Consciousness and Cognition*, 22(1), 8–21. <https://doi.org/10.1016/j.concog.2012.11>